MILITARY SPECIFICATION

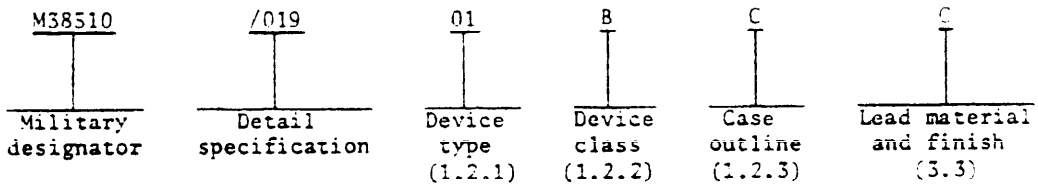
MICROCIRCUITS, DIGITAL
TTL, PARITY GENERATORS/CHECKERS,
MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, parity generators/checkers. Three product assurance classes and a choice of case outline/lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The complete part number shall be as shown in the following example:



1.2.1 Device type. The device type shall be as shown in the following:

Device type	Circuit
01	8-bit odd/even parity generator/checker

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	MIL-M-38510, appendix C, case outline
A	F-1 (14-pin, 1/4" x 1/4" flat pack)
B	F-3 (14-pin, 1/8" x 1/4" flat pack)
C	D-1 (14-pin, 1/4" x 3/4" dual-in-line pack)
D	F-2 (14-pin, 1/4" x 3/8" flat pack)

1.3 Absolute maximum ratings.

Supply voltage range - - - - - - - - - - -0.5 Vdc to 7.0 Vdc
Input voltage range - - - - - - - - - - -1.5 Vdc at -12 mA to 5.5 Vdc
Storage temperature range - - - - - - - -65°C to 150°C
Maximum power dissipation P_D 1/ - - - - 270 mWdc
Lead temperature (soldering 10 seconds) - - - - - - - - - - - - - - 300°C
Thermal resistance, junction to case - - θ_{JC} = 0.09°C/mW for flat pack; 0.08°C/mW for dual-in-line pack
Junction temperature - - - - - - - - - - T_J = 175°C

1/ Must withstand the added P_D due to short circuit condition (e.g., I_{OS}) at one output for 5 seconds duration.

FSC 5962

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1.4 Recommended operating conditions:

Supply voltage	4.5 Vdc minimum to 5.5 Vdc maximum
Minimum high level input voltage	2.0 Vdc
Maximum low level input voltage	0.8 Vdc
Normalized fanout (each output):	
Logical high	20 maximum
Logical low	10 maximum
Ambient operating temperature range	-55°C to 125°C

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Truth tables. The truth tables shall be as specified on figure 2.

3.2.3 Logic diagram. The logic diagram shall be as specified on figure 3.

3.2.4 Schematic circuit. The schematic circuit shall be as specified on figure 4.

3.2.5 Case outlines. The case outlines shall be in accordance with 1.2.3.

3.3 Lead material and finish. Lead material and finish shall be in accordance with MIL-M-38510.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I and apply over the full recommended ambient operating temperature range, unless otherwise specified.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions	Test limits		
			Min	Max	Unit
High level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IH} = 2 V V_{IL} = 0.8 V, I_{OH} = -800 µA	2.4	---	V
Low level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IH} = 2 V V_{IL} = 0.8 V, I_{OL} = 16 mA	---	0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -12 mA T_A = 25°C	---	-1.5	V
High level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	---	40	µA
High level input current	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	---	1	mA
Low level input current	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	-0.7	-1.6	mA
Short circuit output current	I_{OS}	V_{CC} = 5.5 V 1/	-20	-55	mA
Supply current	I_{CC}	V_{CC} = 5.5 V, V_{odd} = V_{even} = 2.4 V	---	49	mA
Propagation delay to high logic level (data to Σ even, even input grounded)	t_{PLH1}	C_L = 50 pF, R_L = 400Ω	2	67	ns

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions	Test limits		
			Min	Max	Unit
Propagation delay to low logic level (data to Σ even, even input grounded)	t_{PHL1}	C_L = 50 pF, R_L = 400Ω	2	54	ns
Propagation delay to high logic level (data to Σ even, odd input grounded)	t_{PLH2}	C_L = 50 pF, R_L = 400Ω	2	83	ns
Propagation delay to low logic level (data to Σ even, odd input grounded)	t_{PHL2}	C_L = 50 pF, R_L = 400Ω	2	96	ns
Propagation delay to high logic level (data to Σ odd, odd input grounded)	t_{PLH3}	C_L = 50 pF, R_L = 400Ω	2	67	ns
Propagatin delay to low logic level (data to Σ odd, odd input grounded)	t_{PHL3}	C_L = 50 pF, R_L = 400Ω	2	54	ns
Propagation delay to low logic level (data to Σ odd, even input grounded)	t_{PLH4}	C_L = 50 pF, R_L = 400Ω	2	83	ns
Propagation delay to low logic level (data to Σ odd, even input grounded)	t_{PHL4}	C_L = 50 pF, R_L = 400Ω	2	93	ns
Propagation delay to high logic level (even or odd to Σ even or Σ odd)	t_{PLH5}	C_L = 50 pF, R_L = 400Ω	2	31	ns
Propagation delay to low logic level (even or odd to Σ even or Σ odd)	t_{PHL5}	C_L = 50 pF, R_L = 400Ω	2	17	ns

1/ Not more than one output should be shorted at a time.

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3.5 Rebonding. Rebonding shall be in accordance with MIL-M-38510.

3.6 Electrical test requirements. Electrical test requirements shall be as specified in table III for the applicable device type and device class. The subgroups of table III which constitute the minimum electrical test requirements for screening, qualification, and quality conformance by device class are specified in table II.

3.7 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2. At the option of the manufacturer, the following marking may be omitted from the body of the microcircuit, but shall be retained on the initial container:

(a) Country of origin.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirement	Subgroups (see table III)		
	Class A devices	Class B devices	Class C devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1	None
Final electrical test parameters (method 5004)	1*, 2, 3, 9, 10, 11	1*, 2, 3, 9	1
Group A test requirements (method 5005)	1, 2, 3, 9 10, 11	1, 2, 3, 9	1, 2, 3, 9
Group C end point electrical parameters (method 5005)	1, 2, 3	1, 2, 3	1
Additional electrical subgroups for group C periodic inspections	None	10, 11	10, 11

*PDA applies to subgroup 1 (see 4.3(h)).

4. PRODUCT ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and method 5005 of MIL-STD-883, except as modified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, and C inspections (see 4.4.1, 4.4.2, and 4.4.3). After qualification of one or more electrically and structurally similar types with a single lead finish, other lead finishes of the same case outline may be qualified by submitting a single type in the qualified case outline to the group B, subgroup 3 test and the group C, subgroups 1, 3, and 4 tests.

4.3 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

(a) Test samples for the group B bond strength test specified in method 5005 of MIL-STD-883 may, at the manufacturer's option, be randomly selected immediately following the internal visual (precap) inspection and prior to sealing (see 4.4.2(a)).

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(b) Burn-in test (method 1015 of MIL-STD-883).
(1) Test condition D or E, using the circuit shown on figure 5, or equivalent.
(2) T_A = 125°C minimum.

(c) Reverse bias burn-in and interim electrical test in accordance with 3.1.10 of method 5004 of MIL-STD-883 may be omitted.

(d) Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

(e) Percent defective allowable (PDA) - The PDA is specified as 5 percent for class A devices and 10 percent for class B devices based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA.

The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510.

4.4.1 Group A inspection. Group A inspection shall consist of the test subgroups and LTPD values shown in table I of method 5005 of MIL-STD-883 and as follows:

(a) Tests shall be as specified in table II.
(b) Subgroups 4, 5, and 6 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall consist of the test subgroups and LTPD values shown in table II of method 5005 of MIL-STD-883 and as follows:

(a) Bond strength test may be conducted on samples collected prior to sealing (see 4.3(a)).

4.4.3 Group C inspection. Group C inspection shall consist of the test subgroups and LTPD values shown in table III of method 5005 of MIL-STD-883 and as follows:

(a) End point electrical parameters shall be as specified in table II.
(b) Subgroups 7 and 8 shall be added to the group C inspection requirements for class B and class C devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.
(c) High temperature storage test (method 1008 of MIL-STD-883) conditions:
(1) Temperature: 150 ±10°C.
(2) Duration: 1,000 hours, except as otherwise permitted by appendix B of MIL-M-38510.
(d) Operating life test (method 1005 of MIL-STD-883) conditions:
(1) Test condition D or E, using the circuit shown on figure 5, or equivalent.
(2) T_A = 125°C, minimum.
(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510.
(e) Omit steady state reverse bias test.

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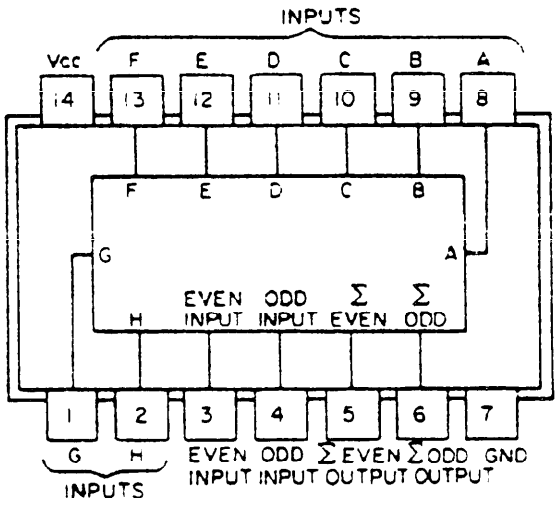


FIGURE 1. Terminal connections (top view).

INPUTS			OUTPUTS	
Σ of H's At A thru H	EVEN	ODD	Σ EVEN	Σ ODD
EVEN	H	L	H	L
ODD	H	L	L	H
EVEN	L	H	L	H
ODD	L	H	H	L
X	H	H	L	L
X	L	L	H	H

X = Irrelevant

FIGURE 2. Truth table.

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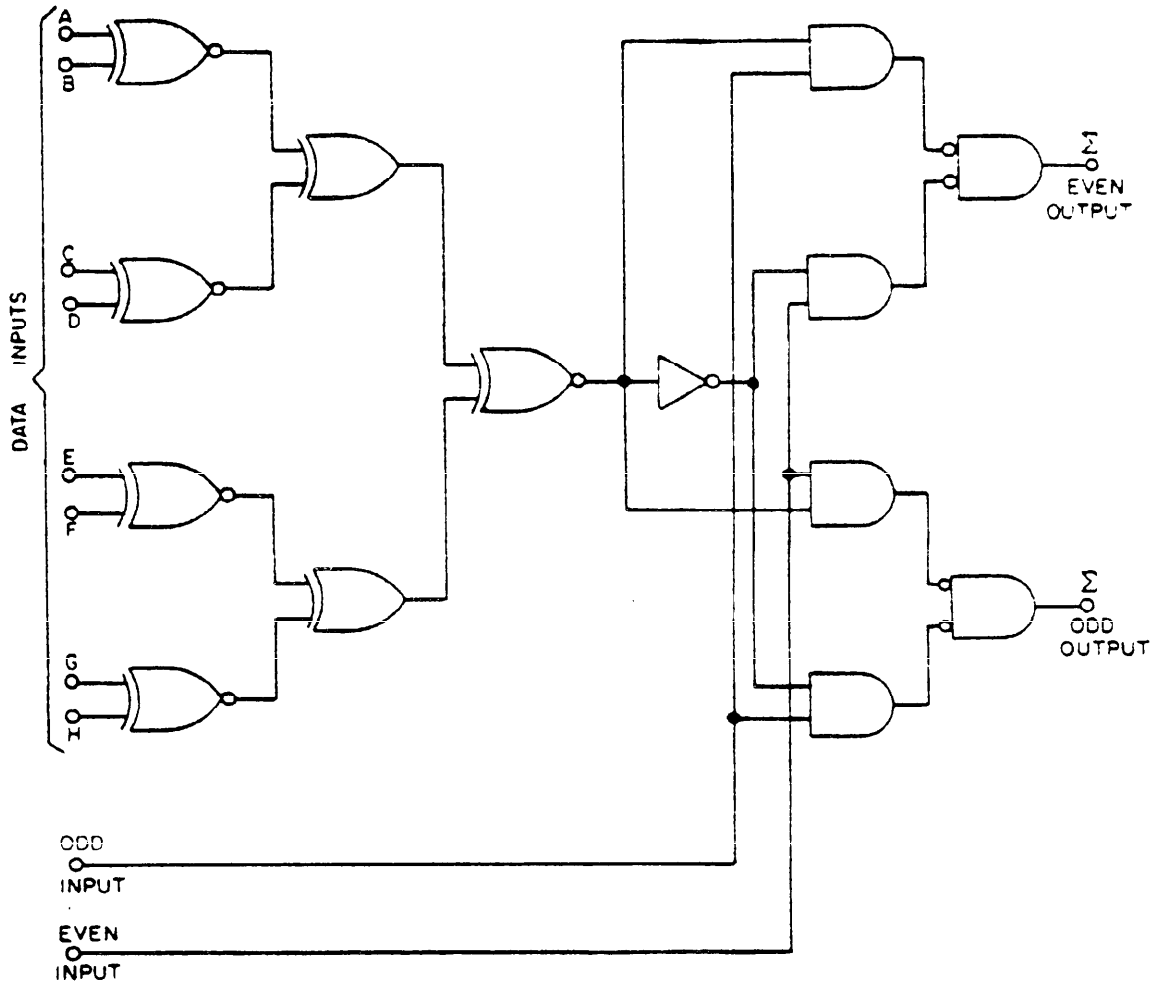


FIGURE 3. Logic diagram.

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Device type 01

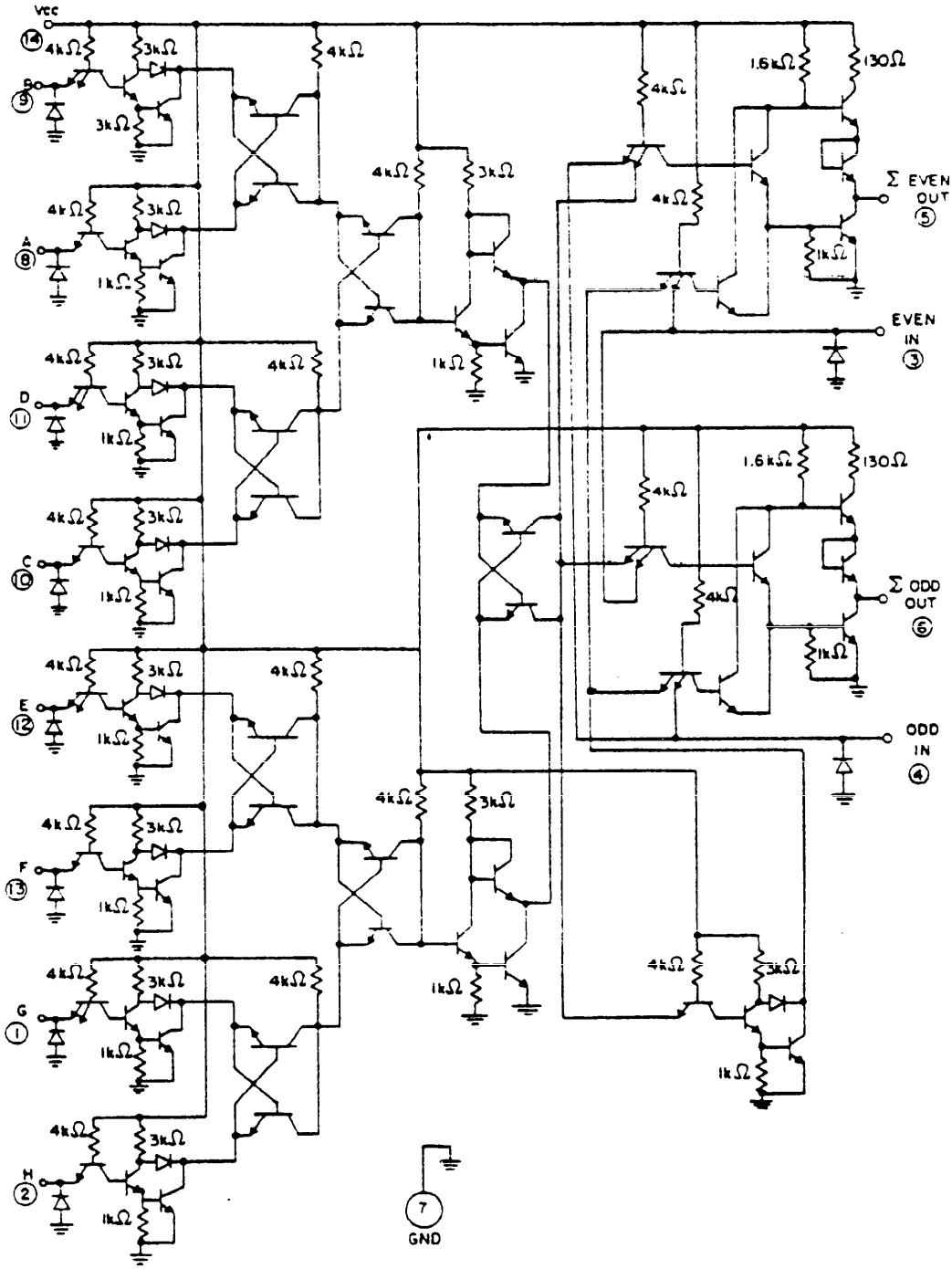
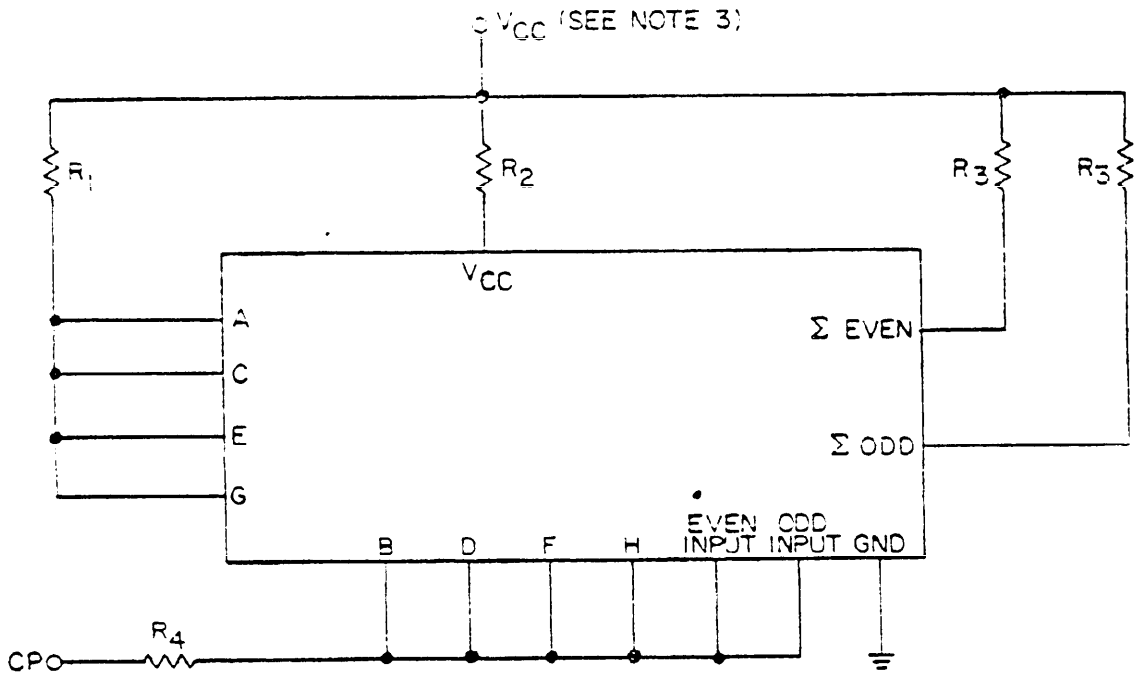


FIGURE 4. Schematic circuit.

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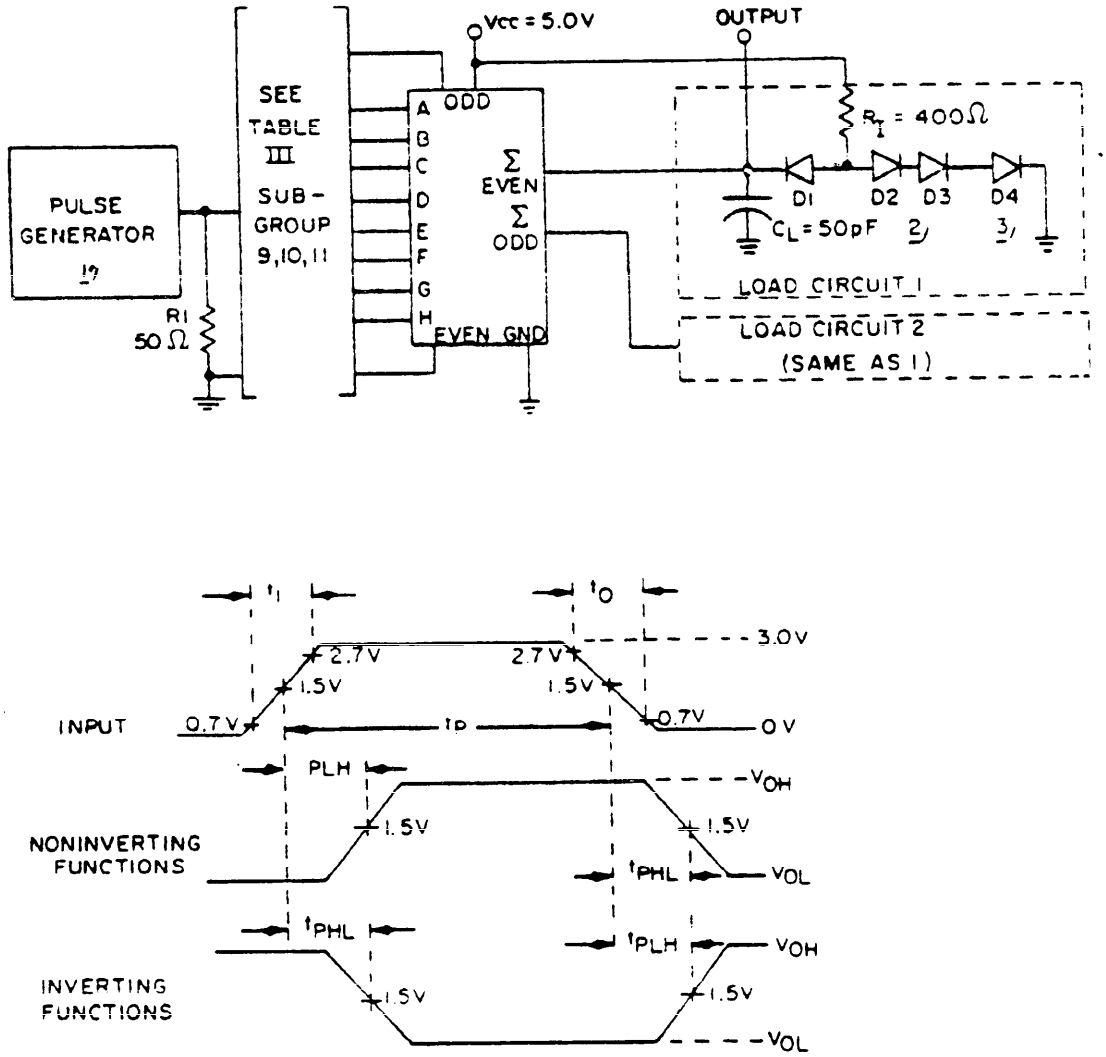
Device type 01



NOTES:
1. R_1 = 1 kΩ ±5%; R_2 = 4.7Ω ±5%; R_3 = 220Ω ±5%; R_4 = 27Ω ±5%.
2. CP = 100 kHz, 0-3 V square wave, 50% duty cycle.
3. Voltage shall be such that a minimum of 5.0 V shall be present at V_{CC} terminal.

FIGURE 5. Burn-in and life test circuit.

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NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3 V, $t_1 = t_0$ = 10 ns, t_p = 500 ns, PRR = 1 MHz, and $Z_{out} \approx 50\Omega$.
2. C_L includes probe and jig capacitance.
3. All diodes are 1N3064 or equivalent.
4. Load circuits for a given output are required only when the specific test in table III indicates "OUT" for that output. Otherwise load circuits may be omitted.

FIGURE 6. Switching time test circuit.

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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ .8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Case A,B,C,D Test No.	1 Inp G	2 Inp H	3 Even IN	4 Odd IN	5 Σ Even OUT	6 Σ Odd OUT	7 GND	8 Inp A	9 Inp B	10 Inp C	11 Inp D	12 Inp E	13 Inp F	14 V_{CC}	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
1	V_{OH}	3006	1	2.0 V	0.8 V	0.8 V	0.8 V	-.8 mA		GND	2.0 V	0.8 V	2.0 V	0.8 V	2.0 V	0.8 V	4.5 V	Σ Even	2.4		Vdc
T_A = 25°C			2		0.8 V	0.8 V	0.8 V		-.8 mA									Σ Odd			
			3		0.8 V	2.0 V	0.8 V	-.8 mA										Σ Even			
			4		2.0 V	2.0 V	0.8 V		-.8 mA									Σ Odd			
			5		0.8 V	0.8 V	2.0 V		-.8 mA									Σ Odd			
			6		2.0 V	0.8 V	2.0 V	-.8 mA										Σ Even			
	V_{OL}	3007	7		0.8 V	2.0 V	2.0 V	16 mA										Σ Even		0.4	
			8		0.8 V	2.0 V	2.0 V		16 mA									Σ Odd			
			9		0.8 V	2.0 V	0.8 V		16 mA									Σ Odd			
			10		2.0 V	2.0 V	0.8 V	16 mA										Σ Even			
			11		0.8 V	0.8 V	2.0 V	16 mA										Σ Even			
			12		2.0 V	0.8 V	2.0 V		16 mA									Σ Odd			
	V_{IC}		13								-12 mA	-12 mA						Inp A		-1.5	
			14										-12 mA					Inp B			
			15											-12 mA				Inp C			
			16												-12 mA			Inp D			
			17													-12 mA		Inp E			
			18															Inp F			
			19	-12 mA														Inp G			
			20		-12 mA													Inp H			
			21			-12 mA												Even IN			
			22				-12 mA											Odd IN			
	I_{IH1}	3010	23								2.4 V						5.5 V	Inp A		40	μA
			24									2.4 V						Inp B			
			25										2.4 V					Inp C			
			26											2.4 V				Inp D			
			27												2.4 V			Inp E			
			28													2.4 V		Inp F			
			29	2.4 V														Inp G			
			30		2.4 V													Inp H			
			31	4.5 V	GND	2.4 V					4.5 V	GND	4.5 V	GND	4.5 V	GND		Even IN			
			32				2.4 V					GND						Odd IN			
			33			2.4 V						4.5 V						Even IN			
			34				2.4 V					4.5 V						Odd IN			
	I_{IH2}		35								5.5 V							Inp A		1	mA
			36									5.5 V						Inp B			
			37										5.5 V					Inp C			
			38											5.5 V				Inp D			
			39												5.5 V			Inp E			
			40													5.5 V		Inp F			
			41	5.5 V														Inp G			
			42		5.5 V													Inp H			
			43	4.5 V	GND	5.5 V					4.5 V	GND	4.5 V	GND	4.5 V	GND		Even IN			
			44				5.5 V					GND						Odd IN			
			45			5.5 V						4.5 V						Even IN			
			46				5.5 V					4.5 V						Odd IN			

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TABLE III. Group A inspection for device type 01 - Continued.

Terminal conditions (pins not designated may be high $\geq$ 2.0 V or low $\leq$ .8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Case A,B,C,D Test No.	1 Inp G	2 Inp H	3 Even IN	4 Odd IN	5 Σ Even OUT	6 Σ Odd OUT	7 GND	8 Inp A	9 Inp B	10 Inp C	11 Inp D	12 Inp E	13 Inp F	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_A 25°C	I_{IL1}	3009	47							GND	0.4 V	0.4 V					5.5 V	Inp A	-0.7	-1.6	mA
			48										0.4 V					Inp B			
			49											0.4 V				Inp C			
			50												0.4 V			Inp D			
			51													0.4 V		Inp E			
			52															Inp F			
			53	0.4 V														Inp G			
			54		0.4 V													Inp H			
			55	2.0 V	GND	0.4 V					2.0 V	GND	2.0 V	GND	2.0 V	GND		Even IN			
			56				0.4 V					GND						Odd IN			
			57			0.4 V						2.0 V						Even IN			
			58				0.4 V					2.0 V						Odd IN			
	I_{OS}	3011	59	0.8 V	2.0 V	GND	GND	GND			2.0 V	0.8 V	2.0 V	0.8 V	0.8 V	2.0 V		Σ Even OUT	-20	-55	
	I_{OS}	3011	60	0.8 V	2.0 V	GND	GND		GND		2.0 V	0.8 V	2.0 V	0.8 V	0.8 V	2.0 V		Σ Odd OUT	-20	-55	
	I_{CC}	3005	61			2.4 V	2.4 V											V_{CC}		49	
2	Same tests, terminal conditions, and limits as subgroup 1, except T_A = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as subgroup 1, except T_A = -55°C and V_{IC} tests are omitted.																				
9 T_A 25°C	t_{PLH1}	3003 (Fig 6)	62	GND	GND	GND	5.0 V	OUT		GND	IN	GND	GND	GND	GND	GND	5.0 V	A to Σ Even	2	52	ns
			63								GND	IN	GND					B to Σ Even			
			64									GND	IN					C to Σ Even			
			65										GND	IN				D to Σ Even			
			66											GND	IN			E to Σ Even			
			67												GND	IN		F to Σ Even			
			68	IN												GND		G to Σ Even			
			69	GND	IN											GND		H to Σ Even			

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ .8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, C, D Test No.	1 Inp G	2 Inp H	3 Even IN	4 Odd IN	5 Σ Even OUT	6 Σ Odd OUT	7 GND	8 Inp A	9 Inp B	10 Inp C	11 Inp D	12 Inp E	13 Inp F	14 V_{CC}	Meas terminal	Test limits Min	Max	Unit
9 T_A = 25°C	t_{PHL1}	3003 (Fig 6)	70	GND	GND	GND	5.0 V	OUT		GND	IN	GND	GND	GND	GND	GND	5.0 V	A to Σ Even	2	42	ns
			71								GND	IN	GND					B to Σ Even			
			72									GND	IN					C to Σ Even			
			73										GND	IN				D to Σ Even			
			74											GND	IN			E to Σ Even			
			75												GND	IN		F to Σ Even			
			76	IN												GND		G to Σ Even			
			77	GND	IN											GND		H to Σ Even			
	t_{PLH2}		78	GND	GND	5.0 V	GND	OUT			IN	GND	GND	GND	GND	GND		A to Σ Even		64	
			79								GND	IN	GND					B to Σ Even			
			80									GND	IN					C to Σ Even			
			81										GND	IN				D to Σ Even			
			82											GND	IN			E to Σ Even			
			83												GND	IN		F to Σ Even			
			84	IN												GND		G to Σ Even			
			85	GND	IN											GND		H to Σ Even			
	t_{PHL2}		86	GND	GND	5.0 V	GND	OUT			IN	GND	GND	GND	GND	GND		A to Σ Even		72	
			87								GND	IN	GND					B to Σ Even			
			88									GND	IN					C to Σ Even			
			89										GND	IN				D to Σ Even			
			90											GND	IN			E to Σ Even			
			91												GND	IN		F to Σ Even			
			92	IN												GND		G to Σ Even			
			93	GND	IN											GND		H to Σ Even			

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TABLE III. Group A inspection for device type 01 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low $\leq .8$ V, or open)

Subgroup	Symbol	MIL-STD-883 method	Case A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Test Limits		
			Test No.	Inp G	Inp H	Even IN	Odd IN	Σ Even OUT	Σ Odd OUT	GND	Inp A	Inp B	Inp C	Inp D	Inp E	Inp F	V_{CC}	Meas. terminal	Min	Max	Unit
9 T_A = 25°C	t_{PLH3}	3003 (Fig 6)	94	GND	GND	5.0 V	GND		OUT	GND	IN	GND	GND	GND	GND	GND	5.0 V	A to Σ Odd	2	52	ns
			95								GND	IN	GND					B to Σ Odd			
			96									GND	IN					C to Σ Odd			
			97										GND	IN				D to Σ Odd			
			98											GND	IN			E to Σ Odd			
			99												GND	IN		F to Σ Odd			
			100	IN												GND		G to Σ Odd			
			101	GND	IN											GND		H to Σ Odd			
	t_{PHL3}		102	GND	GND	5.0 V	GND		OUT		IN	GND	GND	GND	GND	GND		A to Σ Odd		42	
			103								GND	IN	GND					B to Σ Odd			
			104									GND	IN					C to Σ Odd			
			105										GND	IN				D to Σ Odd			
			106											GND	IN			E to Σ Odd			
			107												GND	IN		F to Σ Odd			
			108	IN												GND		G to Σ Odd			
			109	GND	IN											GND		H to Σ Odd			
	t_{PLH4}		110	GND	GND	GND	5.0 V		OUT		IN	GND	GND	GND	GND	GND		A to Σ Odd		64	
			111								GND	IN	GND					B to Σ Odd			
			112									GND	IN					C to Σ Odd			
			113										GND	IN				D to Σ Odd			
			114											GND	IN			E to Σ Odd			
			115												GND	IN		F to Σ Odd			
			116	IN												GND		G to Σ Odd			
			117	GND	IN											GND		H to Σ Odd			

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ .8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Case A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits		
			Test No.	Inp G	Inp H	Even IN	Odd IN	Σ Even OUT	Σ Odd OUT	GND	Inp A	Inp B	Inp C	Inp D	Inp E	Inp F	V_{CC}		Min	Max	Unit
9	t_{PHL4}	3003 (Fig 6)	118	GND	GND	GND	5.0 V		OUT	GND	IN	GND	GND	GND	GND	GND	5.0 V	A to Σ Odd	2	72	ns
T_A = 25°C			119								GND	IN	GND					B to Σ Odd			
			120									GND	IN					C to Σ Odd			
			121										GND	IN				D to Σ Odd			
			122											GND	IN			E to Σ Odd			
			123												GND	IN		F to Σ Odd			
			124	IN												GND		G to Σ Odd			
			125	GND	IN											GND		H to Σ Odd			
	t_{PLH5}		126	5.0 V	5.0 V	IN	GND	OUT			GND	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V		Even IN to Σ Even		24	
			127			IN	GND		OUT		5.0 V							Even IN to Σ Odd			
			128			GND	IN	OUT			5.0 V							Odd IN to Σ Even			
			129			GND	IN		OUT		GND							Odd IN to Σ Odd			
	t_{PHL5}		130	5.0 V	5.0 V	IN	GND	OUT			GND	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V		Even IN to Σ Even		14	
			131			IN	GND		OUT		5.0 V							Even IN to Σ Odd			
			132			GND	IN	OUT			5.0 V							Odd IN to Σ Even			
			133			GND	IN		OUT		GND							Odd IN to Σ Odd			
10	t_{PLH1}	3003 (Fig 6)	134	GND	GND	GND	5.0 V	OUT			IN	GND	GND	GND	GND	GND	5.0 V	A to Σ Even	2	67	ns
T_A = 125°C			135								GND	IN	GND					B to Σ Even			
			136									GND	IN					C to Σ Even			
			137										GND	IN				D to Σ Even			
			138											GND	IN			E to Σ Even			
			139												GND	IN		F to Σ Even			
			140	IN												GND		G to Σ Even			
			141	GND	IN											GND		H to Σ Even			

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ .8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Case A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits		
			Test No.	Inp G	Inp H	Even IN	Odd IN	Σ Even OUT	Σ Odd OUT	GND	Inp A	Inp B	Inp C	Inp D	Inp E	Inp F	V_{CC}		Min	Max	Unit
10 T_A 125°C	t_{PHL1}	3003 (Fig 6)	142	GND	GND	GND	5.0 V	OUT		GND	IN	GND	GND	GND	GND	GND	5.0 V	A to Σ Even	2	54	ns
			143								GND	IN	GND					B to Σ Even			
			144									GND	IN					C to Σ Even			
			145										GND	IN				D to Σ Even			
			146											GND	IN			E to Σ Even			
			147												GND	IN		F to Σ Even			
			148	IN												GND		G to Σ Even			
			149	GND	IN											GND		H to Σ Even			
	t_{PLH2}		150	GND	GND	5.0 V	GND	OUT			IN	GND	GND	GND	GND	GND		A to Σ Even		83	
			151								GND	IN	GND					B to Σ Even			
			152									GND	IN					C to Σ Even			
			153										GND	IN				D to Σ Even			
			154											GND	IN			E to Σ Even			
			155												GND	IN		F to Σ Even			
			156	IN												GND		G to Σ Even			
			157	GND	IN											GND		H to Σ Even			
	t_{PHL2}		158	GND	GND	5.0 V	GND	OUT			IN	GND	GND	GND	GND	GND		A to Σ Even		96	
			159								GND	IN	GND					B to Σ Even			
			160									GND	IN					C to Σ Even			
			161										GND	IN				D to Σ Even			
			162											GND	IN			E to Σ Even			
			163												GND	IN		F to Σ Even			
			164	IN												GND		G to Σ Even			
			165	GND	IN											GND		H to Σ Even			

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ .8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Case A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits		
			Test No.	Inp G	Inp H	Even IN	Odd IN	Σ Even OUT	Σ Odd OUT	GND	Inp A	Inp B	Inp C	Inp D	Inp E	Inp F	V_{CC}		Min	Max	Unit
10 T_A = 125°C	t_{PLH3}	3003 (Fig 6)	166	GND	GND	5.0 V	GND		OUT	GND	IN	GND	GND	GND	GND	GND	5.0 V	A to Σ Odd	2	67	ns
			167								GND	IN	GND					B to Σ Odd			
			168									GND	IN					C to Σ Odd			
			169										GND	IN				D to Σ Odd			
			170											GND	IN			E to Σ Odd			
			171												GND	IN		F to Σ Odd			
			172	IN												GND		G to Σ Odd			
			173	GND	IN											GND		H to Σ Odd			
	t_{PHL3}		174	GND	GND	5.0 V	GND		OUT		IN	GND	GND	GND	GND	GND		A to Σ Odd		54	
			175								GND	IN	GND					B to Σ Odd			
			176									GND	IN					C to Σ Odd			
			177										GND	IN				D to Σ Odd			
			178											GND	IN			E to Σ Odd			
			179												GND	IN		F to Σ Odd			
			180	IN												GND		G to Σ Odd			
			181	GND	IN											GND		H to Σ Odd			
	t_{PLH4}		182	GND	GND	GND	5.0 V		OUT		IN	GND	GND	GND	GND	GND		A to Σ Odd		83	
			183								GND	IN	GND					B to Σ Odd			
			184									GND	IN					C to Σ Odd			
			185										GND	IN				D to Σ Odd			
			186											GND	IN			E to Σ Odd			
			187												GND	IN		F to Σ Odd			
			188	IN												GND		G to Σ Odd			
			189	GND	IN											GND		H to Σ Odd			

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TABLE III. Group A inspection for device type 01 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ .8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Case A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Test limits		
			Test No.	Inp G	Inp H	Even IN	Odd IN	Σ Even OUT	Σ Odd OUT	GND	Inp A	Inp B	Inp C	Inp D	Inp E	Inp F	V_{CC}	Meas terminal	Min	Max	Unit
10 T_A = 125°C	t_{PHL4}	3003 (Fig. 6)	190	GND	GND	GND	5.0 V		OUT	GND	IN	GND	GND	GND	GND	GND	5.0 V	A to Σ Odd	2	93	ns
			191								GND	IN	GND					B to Σ Odd			
			192									GND	IN					C to Σ Odd			
			193										GND	IN				D to Σ Odd			
			194											GND	IN			E to Σ Odd			
			195												GND	IN		F to Σ Odd			
			196	IN												GND		G to Σ Odd			
			197	GND	IN											GND		H to Σ Odd			
	t_{PLH5}		198	5.0 V	5.0 V	IN	GND	OUT			GND	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V		Even IN to Σ Even		31	
			199			IN	GND		OUT		5.0 V							Even IN to Σ Odd			
			200			GND	IN	OUT			5.0 V							Odd IN to Σ Even			
			201			GND	IN		OUT		GND							Odd IN to Σ Odd			
	t_{PHL5}		202	5.0 V	5.0 V	IN	GND	OUT			GND	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V		Even IN to Σ Even		17	
			203			IN	GND		OUT		5.0 V							Even IN to Σ Odd			
			204			GND	IN	OUT			5.0 V							Odd IN to Σ Even			
			205			GND	IN		OUT		GND							Odd IN to Σ Odd			
11	Same tests, terminal conditions and limits as subgroup 10, except T_A = -55°C.																				

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4.5 Methods of examination and test. Methods of examination and test shall be as specified in the appropriate tables and as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Life test cooldown procedure. When devices are measured at 25 C following application of the operating life or burn-in test condition, they shall be cooled to room temperature prior to removal of the bias. Alternately, the bias may be removed during cooling if the case temperature is reduced to room temperature within 30 minutes after removal of the test condition.

4.6 Inspection of preparation for delivery. Inspection of preparation for delivery shall be in accordance with MIL-M-38510, except that the rough handling test shall not apply.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging and packing. Microcircuits shall be prepared for delivery in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for use for Government microcircuit applications (original equipment) and logistic purposes.

6.3 Ordering data. The contract or order should specify the following:

(a) Complete part number (see 1.2).
(b) Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.
(c) Requirement for certificate of compliance, if applicable.
(d) Requirements for notification of change of product or process to procuring activity in addition to notification to the qualifying activity, if applicable.
(e) Requirements for packaging and packing.
(f) Requirements for failure analysis (including required test condition of Method 5003), corrective action and reporting of results, if applicable.
(g) Requirements for product assurance options.
(h) Requirements for carriers, special lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-STD-1313, MIL-STD-1331, and as follows:

GND - - - - - - Electrical ground (common terminal)
V_{IN} - - - - - - Voltage level at an input terminal

6.5 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits procured for Government logistic support will be procured to device class B (see 1.2.2) and lead material and finish C (see 3.3). There is no substitute for case outline C. Longer length leads and lead forming shall not affect the part number.

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6.6 Substitutability. Microcircuits covered by this specification unilaterally supersede the following commercial device type:

Device type	Commercial type
01	54180

Custodian:
Air Force - 17

Review activities:
Air Force - 11, 70, 80
DSA - ES

User activity:
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DSA - ES

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